Sergio **Manthey Corchado**

Particle Physicist · Ph.D. Candidate

CIEMAT / Universidad Autónoma de Madrid

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Academic Education

Universidad Autónoma de Madrid — *Madrid, Spain*

Ph.D. in Physics — *2021 - 2026*

Thesis: Liquid Argon Detector Development for the Neutrino Oscillation Measurement with the DUNE experiment. Thesis submitted; defence pending.

Università di Roma La Sapienza — *Rome, Italy*

M.S. in Particle and Astroparticle Physics — *2018 - 2021*

Thesis: BULLKID. Low Mass Dark Matter Research with Microwave Kinetic Inductance Detectors (MKIDS).

University of Bremen — *Bremen, Germany*

B.S. in Physics — *2014 - 2018*

Thesis: Oxygen-Radical Production through VUV-Light for the Surface Treatment of Adhesive and Lacquer Processes.

Work Experience

Responsibilities

- Lead analyst of the CIEMAT apparatus for the VD **X-ARAPUCA Characterisation**.
- Software implementation of **Solar Neutrino Reconstruction** chain in LArSoft.

Positions

Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas — *Madrid, Spain*

Scientific Researcher — *2021 - 2025*

- Join professional membership of **DUNE**, **ProtoDUNE** & **SoLAr** experiments.
- Perform laboratory measurements of cryogenic **Liquid Argon Scintillation** light detection with photosensors.
- Develop **Monte-Carlo** simulations and reconstruction tools for **Low-Energy Neutrinos** analyses.
- Produce low-energy reconstruction and sensitivity analysis of **Solar Neutrinos**.
- Participate in **Outreach Program** for national secondary school.

Duke University — *Durham, USA*

Visiting researcher at Duke University — *Jul 2023 - Sep 2023*

Develop **Low-Energy Analysis Tools** in LArSoft for solar neutrino detection.

European Space Agency (ESA-ESAC) — *Villafranca del Castillo, Spain*

Visiting research contract from ESA — *Jul 2021 - Sep 2021*

Outreach for scientific project (CesaR).

Ministry of Foreign Affairs and International Cooperation (MAECI) — *Rome, Italy*

Research Under Academic Supervision (Grant) — *Jan 2020 - Jun 2020*

Detector science in the cryogenic laboratory for **Dark Matter R&D**.

Fraunhofer Institute (IFAM) — *Bremen, Germany*

Student Researcher — *Nov 2016 - May 2018*

Laboratory assistant in the department for **Adhesive Bonding and Plasma Technologies**.

Skills

Technical

Programming C++, Python, Geant4, LArSoft and LaTeX

Computing Linux, Git and HTCondor

Language

Spanish Mother tongue. Speaking: native, Reading: native, Writing: native and Listening: native
English Cambridge – C2. Speaking: proficient, Reading: proficient, Writing: proficient and Listening: proficient
German EOI Palma – C1. Speaking: advanced, Reading: advanced, Writing: advanced and Listening: advanced
Italian Bremen University – A2. Speaking: intermediate, Reading: intermediate, Writing: intermediate and Listening: intermediate

Conferences

Lake Louise Winter Institute 2025 *Alberta, Canada*
Talk *02/03/2025 - 08/03/2025*
Low energy and BSM searches in DUNE.

Light Detection In Noble Elements Conference 2024 *Sao Paulo, Brazil*
Talk *26/08/2024 - 28/08/2024*
X-ARAPUCA Photon Detection Efficiency for DUNE Far-Detector Vertical-Drift.

42nd International Conference on High Energy Physics *Prague, Czech Republic*
Poster *17/07/2024 - 24/07/2024*
DUNE's Low Energy Physics Searches.

XXXI International Conference on Neutrino Physics and Astrophysics *Milan, Italy*
Poster *17/06/2024 - 21/06/2024*
DUNE Sensitivity to Solar Neutrinos.

19th International Workshop on Low Temperature Detectors *Virtual*
Poster *19/07/2021 - 30/07/2021*
Status of BULLKID, an array of kinetic inductance detectors to search for Dark Matter.

Schools

Dark Matter and Neutrinos 2025 *Paris, France*
Talk *05/05/2025 - 09/05/2025*
DUNE's Sensitivity to Solar Neutrinos.

Research Activities

Deep Underground Neutrino Experiment

Madrid, Spain

PH.D. RESEARCH

2021 - 2025

- My research focuses on advancing the physics capabilities of the **Deep Underground Neutrino Experiment** (DUNE) through both hardware development and data analysis, with a particular emphasis on improving low-energy neutrino detection. My work experience has led me to perform cryogenic laboratory measurements on photosensitive devices, conduct Monte Carlo simulations, and develop reconstruction algorithms for solar neutrinos.

- A central component of my research has been the characterisation and optimisation of **X-ARAPUCA** photon detectors. These devices are a critical part of DUNE's **Photon Detection System** (PDS), responsible for capturing scintillation light produced in DUNE. Because the performance of the PDS directly affects event reconstruction, especially for low-energy interactions such as solar neutrinos, it is essential to characterise its behaviour. My work involves operating these devices in liquid nitrogen or liquid argon environments, triggering on specific events, calibrating a light source and quantifying the system's response. Through these studies, I have computed the expected **Photon Detection Efficiency** for full-scale DUNE X-ARAPUCAs and provided essential input for simulation tuning and detector optimisation. My results have been published in arXiv:2511.12328.

- Bridging the hardware and physics-analysis components of my work, I have contributed extensively to DUNE's **Monte Carlo** simulation framework, focusing on the reconstruction of low-energy charge and scintillation signals. My contributions include implementing waveform filtering and deconvolution techniques, photon pulse reconstruction, event clustering and charge-light matching for further downstream physics analyses. These efforts have been incorporated into the broader **LArSoft Simulation** workflow. A major scientific goal is the reconstruction and study of low-energy neutrinos from the Sun. DUNE's unprecedented combination of mass and high cross-section will allow it to perform precision measurements of the solar neutrino flux and probe matter effects. My research contributes to this effort through the development of vertex and energy reconstruction algorithms that combine scintillation and ionisation signals, and studying event-topology features that distinguish low-energy neutrino interactions from radiological backgrounds.

- Using these tools, I have performed analyses aimed at quantifying DUNE's solar neutrino significance, such as the **Day-Night Asymmetry**, the **HEP Neutrino Discovery** and DUNE's **Sensitivity to Solar Oscillation** parameters. My work also explores how improved photon detection translates into better energy resolution and ultimately stronger physics reach. Results from this work have been shared in doi:10.22323/1.476.0217.

- In summary, my research integrates detector development, detailed simulation, and physics analysis to enhance DUNE's ability to observe and characterise low-energy neutrinos. By improving the performance and understanding of the photon detection system, refining simulation tools, and advancing reconstruction and oscillation analysis techniques, I aim to support DUNE's long-term goal of delivering transformative neutrino physics.

BullKid

Rome, Italy

MASTER RESEARCH

2020 - 2021

- I contributed to the **BULky and Low-Threshold Kinetic Inductance Detectors** (BULLKID) R&D project, focused on the development of cryogenic **Kinetic Inductance Detectors** (KIDs) for the search of rare low-energy processes such as **Low-Mass Dark Matter** and neutrino coherent scattering. I participated in the lithographic design of these MKIDs, tuning their resonance frequencies to achieve multiplexing capabilities. I also performed data quality and noise performance studies on one of the project's first functional prototypes. I produced exclusion curves for low-mass dark matter that could potentially be achieved with a scaled version of this prototype. This work was recognised through my poster presentation at the **Low Temperature Detectors** conference.

Publications

SELECTED ARTICLES AND PROCEEDINGS

*Barthelet; **S. Manthey Corchado** et al.*, “Modeling light signals using data from the first pulsed neutron source program at the DUNE vertical drift ColdBox test facility at the CERN Neutrino Platform”, *JINST* 21(04), 2026. *A. Paudel; W. Shi; P. Sala; F. Cavanna; W. Johnson; J. Wang; W. Ketchum; F. Resnati; A. Heindel; A. Ashkenazi; E.*

doi:10.1088/1748-0221/21/04/P04037 *N. Anfimov; A. Branca; J. Bürgi; L. Calivers; P. Carniti; E. Calvo; E. Cristaldo; C. Cuesta; F. Declich; R. Diurba; P. Dunne; **S. Manthey Corchado*** et al., “Cosmic ray measurements using charge and light readout in a pixelated liquid argon time projection chamber”, *JINST* 21(04), 2026.

doi:10.1088/1748-0221/21/04/P04008 *F. Acerbi; M. Andreotti; A. Balboni; E. Bertolini; S. Bertolucci; G. Botogoske; F. Bramati; A. Branca; C. Brizzolari; G. Brunetti; R. Calabrese; **S. Manthey*** et al., “Cryogenic characterization of FBK NUV-HD-Cryo 3T SiPM sensors for the DUNE photon detection system”, *JINST* 21(02), 2026.

doi:10.1088/1748-0221/21/02/P02047 *G. Botogoske; F. Bruni; E. Calvo; R. Calabrese; N. Canci; A. Canto; C. Cattadori; A. C. Villanueva; S. Coleman; J. Crespo-Anadón; C. Cuesta; **S. Manthey Corchado*** et al., “Measurement of the X-ARAPUCA’s absolute photon detection efficiency for the deep underground neutrino experiment’s vertical drift far detector”, *Eur.Phys.J.C* 86(8), 971, 2026.

doi:10.1140/epjc/s10052-026-16190-1 ***S. Manthey Corchado***, “Measurement of the absolute photon detection efficiency of the DUNE far detector vertical drift X-ARAPUCAs”, *JINST* 20(04), 2025.

doi:10.1088/1748-0221/20/04/C04033 ***S. Manthey Corchado***, “Dune’s low Energy Physics Searches”, *PoS* ICHEP2024, 217, 2025.

doi:10.22323/1.476.0217 *N. Anfimov; A. Branca; J. Bürgi; L. Calivers; C. Cuesta; R. Diurba; P. Dunne; D. Dwyer; J. Evans; A. Ezeribe; A. Gauch; **S. Manthey Corchado*** et al., “First demonstration of a combined light and charge pixel readout on the anode plane of a LArTPC”, *JINST* 19(11), 2024.

doi:10.1088/1748-0221/19/11/P11010 *R. Álvarez-Garrote; C. Brizzolari; A. Canto; E. Calvo; C. Cattadori; C. Cuesta; A. de la Torre Rojo; I. Gil-Botella; C. Gotti; D. Guffanti; A. Machado; **S. Manthey Corchado*** et al., “Measurement of the absolute efficiency of the X-ARAPUCA photon detector for the DUNE Far Detector 1”, *Eur.Phys.J.C* 84(10), 1004, 2024.

doi:10.1140/epjc/s10052-024-13393-2 *R. Álvarez-Garrote; E. Calvo; A. Canto; J. I. Crespo-Anadón; C. Cuesta; A. d. l. T. Rojo; I. Gil-Botella; **S. Manthey Corchado**; I. Martín; C. Palomares; L. Pérez-Molina; A. V. de Osa*, “Measurement of the photon detection efficiency of Hamamatsu VUV4 SiPMS at cryogenic temperature”, *Nucl.Instrum.Meth.A* 1064, 2024.

doi:10.1016/j.nima.2024.169347 *M. Andreotti; S. Bertolucci; A. Branca; C. Brizzolari; G. Brunetti; R. Calabrese; P. Carniti; D. Casazza; C. Cattadori; E. Calvo; A. Cervera; **S. Manthey*** et al., “Cryogenic characterization of Hamamatsu HWB MPPCs for the DUNE photon detection system”, *JINST* 19(01), 2024.

doi:10.1088/1748-0221/19/01/T01007 *S. Parsa; M. Weber; C. Cuesta; I. Gil-Botella; **S. Manthey**; A. M. Szelc; S. W. Li; M. Pallavicini; J. Evans; R. Guenette; D. Marsden; N. McConkey* et al., “SoLAr: Solar Neutrinos in Liquid Argon”, *preprint*, 2022

arXiv:2203.07501

arXiv:2608.04059 *DUNE Collaboration*, “Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber”, *preprint*, 2026

AS A MEMBER OF THE DUNE COLLABORATION (24 PAPERS)

DUNE Collaboration, “A Bayesian approach to the long-baseline neutrino oscillation sensitivity of DUNE”, *preprint*, 2026

arXiv:2607.15927 *DUNE Collaboration*, “Charge readout electronics for the DUNE horizontal drift far detector: design and performance in ProtoDUNE-HD”, *JINST* 21(08), 2026.

doi:10.1088/1748-0221/21/08/P08018 *DUNE Collaboration*, “Reconstruction of atmospheric neutrinos in DUNE’s horizontal-drift far-detector module”, *Eur.Phys.J.C* 86(7), 904, 2026.

doi:10.1140/epjc/s10052-026-15911-w *DUNE Collaboration*, “Measurement of exclusive π+-argon interactions using ProtoDUNE-SP”, *Phys.Rev.D* 113(11), 2026.

doi:10.1103/nm5s-4cnw *DUNE Collaboration*, “First Measurement of π^+-Ar and p-Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data”, *preprint*, 2025

arXiv:2511.11925 *DUNE Collaboration*, “Identification of low-energy kaons in the ProtoDUNE-SP detector”, *Phys.Rev.D* 113(5), 2026.

doi:10.1103/q21l-pl7s *DUNE Collaboration*, “Toward virtual thin-beam \(\nu\) cross-section measurements: A feasibility study of \(\nu\)-Ar interaction analysis with DUNE-PRISM”, *Phys.Rev.D* 114(1), 2026.

doi:10.1103/y9tz-sgcs *DUNE Collaboration*, “Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam”, *Instruments* 10(1), 18, 2026.

doi:10.3390/instruments10010018 *DUNE Collaboration*, “Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP”, *JINST* 20(09), 2025.

doi:10.1088/1748-0221/20/09/P09008 *DUNE Collaboration*, “European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment”, *preprint*, 2025

arXiv:2503.23744 *DUNE Collaboration*, “DUNE Software and Computing Research and Development”, *preprint*, 2025

arXiv:2503.23743 *DUNE Collaboration*, “The DUNE Science Program”, *preprint*, 2025

arXiv:2503.23291 *DUNE Collaboration*, “The DUNE Phase II Detectors”, *preprint*, 2025

arXiv:2503.23293 *DUNE Collaboration*, "Neutrino interaction vertex reconstruction in DUNE with Pandora deep learning", *Eur.Phys.J.C* 85(697), 697, 2025.

doi:10.1140/epjc/s10052-025-14313-8 *DUNE Collaboration*, "The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data", *JINST* 20(02), 2025.

doi:10.1088/1748-0221/20/02/P02021 *DUNE Collaboration*, "DUNE Phase II: scientific opportunities, detector concepts, technological solutions", *JINST* 19(12), 2024.

doi:10.1088/1748-0221/19/12/P12005 *DUNE Collaboration*, "First measurement of the total inelastic cross section of positively charged kaons on argon at energies between 5.0 and 7.5 GeV", *Phys.Rev.D* 110(9), 2024.

doi:10.1103/PhysRevD.110.092011 *DUNE Collaboration*, "Supernova pointing capabilities of DUNE", *Phys.Rev.D* 111(9), 2025.

doi:10.1103/PhysRevD.111.092006 *DUNE Collaboration*, "Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber", *Instruments* 8(3), 2024.

doi:10.3390/instruments8030041 *DUNE Collaboration*, "Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light", *JINST* 19(08), 2024.

doi:10.1088/1748-0221/19/08/P08005 *DUNE Collaboration*, "The DUNE Far Detector Vertical Drift Technology. Technical Design Report", *JINST* 19(08), 2024.

doi:10.1088/1748-0221/19/08/T08004 *DUNE Collaboration*, "Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment", *Phys.Rev.D* 107(11), 2023.

doi:10.1103/PhysRevD.107.112012 *D. Caratelli* et al., "Low-Energy Physics in Neutrino LArTPCs", *preprint*, 2022

doi:10.2172/1861356